

Seminar 4

Problems

- Let m and n be coprime positive integers and $\Phi: \mathbb{Z}_{mn} \rightarrow \mathbb{Z}_m \times \mathbb{Z}_n$ be the isomorphism given by $k \mapsto (k \bmod m, k \bmod n)$. Find a formula for the inverse isomorphism.
- Check if the following groups are isomorphic
 - $\mathbb{Z}_3 \times \mathbb{Z}_6$ and $\mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_3$.
 - $\mathbb{Z}_3 \times \mathbb{Z}_6$ and $\mathbb{Z}_2 \times \mathbb{Z}_9$.
- Write down formulas for an isomorphisms and their inverses:
 - $\mathbb{Z}_{10} \times \mathbb{Z}_3 \simeq \mathbb{Z}_5 \times \mathbb{Z}_6$.
 - $\mathbb{Z}_{30} \simeq \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_5$.
- Find all generators of $\mathbb{Z}_9^*$.
- Write explicitly the formulas for the following sequence of isomorphisms:

$$\mathbb{Z}_{18}^* \simeq \mathbb{Z}_2^* \times \mathbb{Z}_9^* \simeq \mathbb{Z}_9^* \simeq \mathbb{Z}_6$$

- For an element $a \in \mathbb{Z}_n^*$ find its inverse using the fact that $(a, n) = 1$.
- Suppose $G = \mathbb{Z}_n \times \mathbb{Z}_m$ such that m and n are coprime. Show that an element $g = (a, b) \in G$ is a generator of G if and only if a and b are generators in $\mathbb{Z}_n$ and $\mathbb{Z}_m$, respectively.
- Let G be a finite group of order $n = p_1^{k_1} \cdot \dots \cdot p_r^{k_r}$, where $p_1, \dots, p_r$ are distinct prime numbers. Show that an element $g \in G$ is a generator of G if and only if $g^{\frac{n}{p_i}} \neq 1$ for all $1 \leq i \leq r$.
- Find all generators of $\mathbb{Z}_5^*$.
- Find all generators of $\mathbb{Z}_{25}^*$.
- Let G be $\mathbb{Z}_5^*$, $g = 2$. We will use the following table

00	01	10	11
1	2	3	4

to map sequences of 0 and 1 into elements of G . Your lover's public key s is 3. You chose a to be 2.

- Compute your public key $r = g^a = 2^2$.
 - Compute the secret key $k = g^{ab}$.
 - Your lover sent you the message: 2, 3, 2, 1. Recover the original message.
- Let G be $\mathbb{Z}_{17}^*$ and $g = 3$. Suppose $F_4 = \{(a_0, a_1, a_2, a_3) \mid a_i \in \{0, 1\}\}$.
 - Show that the map $\pi: F_4 \rightarrow \mathbb{Z}_{17}^*$ by the rule $(a_0, a_1, a_2, a_3) \mapsto 1 + a_0 + 2a_1 + 4a_2 + 8a_3$ is a bijection.
 - Let $a = 5$ and $b = 3$, compute the public keys $r = g^a$ and $s = g^b$.
 - Compute r^b and s^a and check that the results are the same.
 - Take the sequence $s = (1, 1, 0, 1, 0, 1, 1, 0)$ and cut it in two peaces s_1 and s_2 of the length 4. Then map each peace into $h_1, h_2 \in G$ and encrypt them by the k to get m_1 and m_2 .
 - Decrypt the elements m_1 and m_2 in order to recover the original elements h_1 and h_2 and obtain the original sequence s .
 - Let G be $\mathbb{Z}_{17}^*$ and $g = 5$, $r = g^a = 6$, $s = g^b = 14$, $k = g^{ab}$, and $m = hk = 4$. Try to decrypt the message and recover h by the brute force.

Homework

1. Write down formulas for an isomorphism and its inverse in case $\mathbb{Z}_{10} \times \mathbb{Z}_{12} \simeq \mathbb{Z}_{20} \times \mathbb{Z}_6$.
2. Find all generators of the group $\mathbb{Z}_{49}^*$.
3. Let G be $\mathbb{Z}_{29}^*$ with $g = 2$ being its generator. The size of the group G is enough to encode the English alphabet, space and period. We will use the following table

a	b	c	d	e	f	g	h	i	j	k	l	m	n
1	2	3	4	5	6	7	8	9	10	11	12	13	14
o	p	q	r	s	t	u	v	w	x	y	z		.
15	16	17	18	19	20	21	22	23	24	25	26	27	28

Your lover sent you the public key $s = g^b = 12$. You chose a to be 10.

- (a) Compute the private key $k = g^{ab}$.
 - (b) Decrypt the message from your lover 20, 2, 6, 28, 15, 9, 2, 4, 14, 8.
4. Let G be $\mathbb{Z}_{49}^*$ with $g = 3$ being its generator. The size of the group G is enough to encode the English alphabet. We will use the following table

a	b	c	d	e	f	g	h	i	j	k	l	m
2	4	6	9	11	13	16	18	20	23	25	27	30
n	o	p	q	r	s	t	u	v	w	x	y	z
32	34	37	39	41	44	46	48	3	17	31	45	24

Your lover chose the secret number b to be 7 and you chose your secret number a to be 5.

- (a) Compute your public key $r = g^a$.
- (b) Compute the public key $s = g^b$ of your lover.
- (c) Compute the private key $k = g^{ab}$ and its inverse.
- (d) Decrypt the message from your lover 31, 37, 3, 13, 44, 22.